

# Cross-Lingual Robustness of Tacit Coordination in LLMs: A Human-Referenced Diagnostic Study

## Group Members

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## Project Objective

We aim to build a human-referenced diagnostic benchmark pipeline to quantify whether LLM focal-point behavior in pure coordination (tacit coordination) tasks is robust across languages. Specifically, we will:

- measure LLM–human alignment on coordination items using human response distributions,
- measure cross-lingual stability by comparing LLM output distributions under matched English vs. Chinese prompts,
- deliver a reproducible evaluation pipeline (data wrapper, normalization, metrics, and plots)

## Motivation and Challenges

Recent work has shown that LLMs can coordinate in tacit coordination games and that prompting variants can substantially affect coordination outcomes (Aharon et al., 2026). However, it remains unclear whether LLM focal-point behavior is stable across languages under the same coordination objective. This matters for multilingual deployment and for the reliability of LLM evaluation results. We propose a diagnostic study comparing LLM behavior across English and Chinese prompts on human pure coordination game items, while also measuring alignment with human response distributions.

Key challenges we anticipate are as follows. First, translation and wording differences may shift focal points even when the coordination objective is unchanged. Second, cross-lingual comparisons can be confounded by instruction language vs. answer language; to isolate instruction effects, we will hold the required output language fixed across English and Chinese prompts. Third, open-ended answers require careful normalization (e.g., *New York* / *NYC* / *纽约* to a canonical form), and distribution estimates depend on sampling and decoding settings, which we will control and report for reproducibility.

## Research Questions

- RQ1 (Human Alignment): How closely do LLM responses (English prompts) match human response distributions on pure coordination items?
- RQ2 (Cross-Lingual Stability): For the same coordination task and items, how stable are LLM focal-point distributions across English vs. Chinese prompts when the required output language is held fixed?
- RQ3 (Exploratory: Second-Round Re-Coordination): For items that show a first-round focal mismatch, can a lightweight second-round re-coordination prompt improve alignment or reduce cross-lingual divergence without changing the coordination objective or revealing any human distribution information?

## Experimental Dataset Description

We will use items and human response distributions from *Three International Studies on Pure Coordination Games* (Perez-Zapata et al., 2026). We will focus on a manageable subset of 15–20 items and keep the original coordination framing (i.e., participants answer with the goal of matching another responder). Our dataset setup and preprocessing will include:

- selecting 15–20 items with different levels of human consensus,
- creating English prompts and Chinese prompts that express the same coordination objective,
- adding an explicit answer-language constraint (e.g., “Answer in English”) to enable controlled cross-lingual comparison,
- normalizing answers with simple rule-based canonicalization and manually reviewing ambiguous cases when needed.

## Potential Solutions

We will build a small reproducible evaluation pipeline:

1. Dataset wrapper for item prompts and human distributions
2. Prompt templates in English and Chinese with the same coordination objective (plus the same answer-language constraint)
3. LLM sampling (multiple runs per item/language to estimate answer distributions, with fixed decoding settings and sampling budget)
4. Answer normalization (e.g., **New York** / **NYC** / **纽约** → canonical form)
5. Metric computation and statistical analysis for alignment and cross-lingual stability (including simple uncertainty reporting where applicable)

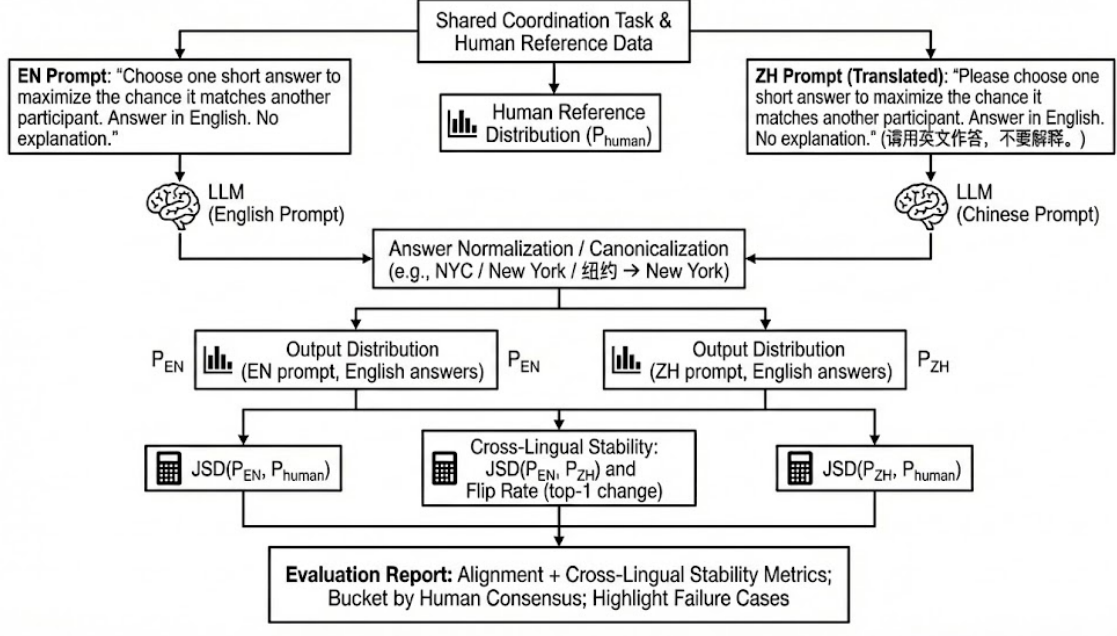


Figure 1: Overview of the evaluation pipeline.

## Evaluation Plan (Metrics and Analysis)

We will evaluate results at the item level and then summarize across items. We will also report uncertainty for key metrics (e.g., JSD and flip rate) using simple bootstrap confidence intervals over repeated samples.

### A. Cross-Lingual Consistency (LLM internal consistency)

- Item-level top-1 match: whether the top response under English and Chinese is the same (0/1)
- Distribution similarity (main metric): similarity between English and Chinese LLM output distributions for the same item
  - Primary metric: JSD (Jensen–Shannon Divergence)
  - Optional secondary metric: TVD (Total Variation Distance)

### B. Human Alignment (LLM-human alignment)

- JSD(LLM, Human) as the main alignment metric (per item)
- Optional:
  - Top-1 human focal match (whether LLM top-1 equals human top-1)
  - Spearman correlation over ranked answer frequencies (if normalization quality is sufficient)

### C. Drift Diagnostics

- Focal-point flip rate: whether the top response changes across English vs. Chinese

- Bucket-wise analysis: group items by human consensus strength (e.g., high / medium / low consensus) and compare:
  - top-1 match
  - JSD
  - flip rate

## D. Exploratory Second-Round Re-Coordination

To incorporate a second-round coordination intuition without introducing a full two-agent interaction protocol, we define first-round “failure” at the focal-point level.

**Definition of first-round mismatch** An item is marked as a first-round mismatch if there is a top-1 mismatch under at least one of the following criteria:

- Type 1 (cross-lingual mismatch): EN top-1  $\neq$  ZH top-1
- Type 2 (human mismatch, optional): LLM top-1  $\neq$  Human top-1

**Round-2 protocol (exploratory)** For mismatch items only, we apply a round-2 prompt that keeps the coordination objective unchanged but provides lightweight feedback (e.g., asking the model to try again to maximize matching/coordination under the same task framing). We will provide no human distribution information and keep the same answer-language constraint as in round 1.

**Round-1 vs. Round-2 comparison** We will compare the following metrics before and after the round-2 prompt:

- top-1 match
- JSD
- focal-point flip rate

In summary, we measure success by (i) alignment to human response distributions and (ii) cross-lingual stability between English vs. Chinese prompts, quantified by JSD/TVD, top-1 agreement, and flip rate (with bootstrap CIs).

## Expected Contribution

This project does not aim to re-test whether prompting can change tacit coordination outcomes. Instead, it provides a human-referenced, cross-lingual robustness diagnostic for tacit coordination in LLMs, along with a reproducible evaluation pipeline (data wrapper, normalization, metrics, and plots) for multilingual analysis. Deliverables will include EN/ZH prompt templates for the selected items, an answer normalization protocol, and a reproducible evaluation script producing all reported metrics and plots.

## References

- Aharon, I., La Malfa, E., Wooldridge, M., & Kraus, S. (2026). *Tacit Coordination of Large Language Models*. arXiv:2601.22184. <https://arxiv.org/abs/2601.22184>
- Perez-Zapata, D., Isoni, A., Zawidzki, T., & Apperly, I. (2026). *Three International Studies on Pure Coordination Games: Adaptable Solutions When Intuitions are Presumed to Vary*. *Journal of Experimental Psychology: General*, 155(3), 760–775. <https://doi.org/10.1037/xge0001876>.